A 2-superirreducible quintic over the rationals

GPT-6 Astra

Abstract

We prove that

$$f(x) = x^5 + 6x^4 - x^3 - 10x^2 - 6x - 1$$

remains irreducible over the rationals after every rational polynomial substitution of degree one or two. This gives a positive answer to the existence question for 2-superirreducible quintics. The proof converts reducibility into a square condition in a quintic field, then into a rational-coordinate condition on an elliptic curve over that field. Unconditional descent, saturation at 3 and 5, and elliptic Chabauty at 3 show that the only relevant elliptic points lie above infinity. The complete Magma verification and an independent exact-arithmetic check of the algebraic identities are supplied.

1 Statement and context

An irreducible polynomial $f \in \mathbb{Q}[x]$ is *2-superirreducible over* $\mathbb{Q}$ if $f(g(x))$ is irreducible in $\mathbb{Q}[x]$ for every $g \in \mathbb{Q}[x]$ with $1 \le \deg g \le 2$. Du [1] constructs several even-degree families and records the existence question in odd degree. The first undecided odd degree in that discussion is five; see also [2].

Theorem 1. *The polynomial*

$$f(x) = x^5 + 6x^4 - x^3 - 10x^2 - 6x - 1 \tag{1}$$

is 2-superirreducible over $\mathbb{Q}$.

The proof is computer-assisted. All field transformations are explicit. The global arithmetic calculation uses Magma's unconditional Mordell–Weil descent, saturation, and elliptic Chabauty routines. Its subgroup-index condition is verified separately and is part of the proof. The coefficient domain throughout is $\mathbb{Q}$, so the assertion covers arbitrary rational quadratic substitutions.

2 A square criterion and an elliptic reformulation

Lemma 1. *Let $F \in \mathbb{Q}[x]$ be monic and irreducible of odd degree $n \ge 3$, let α be a root, and put $K = \mathbb{Q}(\alpha)$. The following conditions are equivalent:*

1. *$F(g(x))$ is reducible for some quadratic $g \in \mathbb{Q}[x]$;*
2. *$\delta^2 = B + A\alpha$ for some $\delta \in K$ and $A, B \in \mathbb{Q}$ with $A \neq 0$;*
3. *there are $t \in \mathbb{Q}$ and $v \in K$ satisfying*

$$v^2 = \frac{F(t)}{t - \alpha}. \tag{2}$$

Proof. Write $g(x) = ax^2 + bx + c$, with $a \neq 0$. If β is a root of $g(x) - \alpha$, then $\alpha = g(\beta)$, so $K \subseteq \mathbb{Q}(\beta)$. The tower formula shows that $F(g(x))$, of degree $2n$, is irreducible exactly when $g(x) - \alpha$ is irreducible over K. The latter quadratic is reducible exactly when its discriminant $(b^2 - 4ac) + 4a\alpha$ is a square in K. Conversely, a square in condition 2 gives the bad substitution $g(x) = (x^2 - B)/A$. This proves the first equivalence.

Assume condition 2 and put $t = -B/A$ and $N = N_{K/\mathbb{Q}}(\delta)$. Since n is odd,

$$N^2 = -A^n F(t).$$

For $n = 2m + 1$, the element $v = N/(A^m \delta)$ consequently satisfies (2). All denominators are nonzero: F has no rational root and $A \neq 0$. Conversely, from (2) one obtains

$$\delta = F(t)/v, \qquad \delta^2 = F(t)(t - \alpha),$$

which is condition 2 with $A = -F(t)$ and $B = tF(t)$. □

For $n = 5$, the polynomial $F(t)/(t - \alpha)$ over K is a squarefree monic quartic. Its smooth projective curve has genus one and two K-rational points above $t = \infty$. Thus the quintic problem becomes an elliptic rational-coordinate problem.

3 The field and the explicit elliptic curve

Put

$$h(x) = x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1, \qquad K = \mathbb{Q}(r), \quad h(r) = 0, \qquad \alpha = r - r^2 - r^3.$$

Both h and f are irreducible modulo 2. Indeed, their reductions are $x^5 + x^4 + x^2 + x + 1$ and $x^5 + x^3 + 1$; neither has a root in $\mathbb{F}_2$ or is divisible by the sole irreducible quadratic $x^2 + x + 1$. A reducible polynomial of degree five has a factor of degree at most two. Direct reduction modulo h gives $f(\alpha) = 0$. Hence α generates K and has minimal polynomial f.

Define the following elements of K:

$$\begin{aligned} k &= r - 1, & b &= -2r^4 - 4r^3 + 6r^2 + 11r - 4, \\ A &= 2r^4 - r^3 - 8r^2 + 2r + 4, \\ B &= 5r^4 + 4r^3 - r^2 - 6r - 41, \\ C &= -11r^4 + 77r^2 - 132. \end{aligned}$$

Consider

$$E: \quad Y^2 = X^3 + AX^2 + BX + C \tag{3}$$

and the function

$$\tau(X, Y) = \frac{X}{kX + b}. \tag{4}$$

The letter b in this section denotes the displayed field element. Let $q(t) = f(t)/(t - \alpha)$. Expansion and reduction modulo h give

$$(kX + b)^4 q\left(\frac{X}{kX + b}\right) = b^2(X^3 + AX^2 + BX + C), \qquad b^2 = 11(4 - r^2). \tag{5}$$

In particular, $b \neq 0$. The maps on the affine charts are

$$t = \frac{X}{kX + b}, \quad v = \frac{bY}{(kX + b)^2}; \qquad X = \frac{bt}{1 - kt}, \quad Y = \frac{bv}{(1 - kt)^2}.$$

For rational t, the denominator $1 - kt$ never vanishes, since $k = r - 1$ is irrational. Every point relevant to Lemma 1 therefore gives a point on E with finite rational τ.

The function τ extends regularly to the point at infinity O of E, with $\tau(O) = 1/k$, which is irrational. Its two poles are

$$\left(-\frac{b}{k}, \frac{b}{k^2}\right), \qquad \left(-\frac{b}{k}, -\frac{b}{k^2}\right). \tag{6}$$

These statements also resolve the removable base point in the projective formula $[X : Y : Z] \mapsto [X : kX + bZ]$.

4 The arithmetic point classification

Proposition 1. *The only points $Q \in E(K)$ for which $\tau(Q) \in \mathbb{P}^1(\mathbb{Q})$ are the two points in (6). Both have $\tau(Q) = \infty$.*

Computer-assisted proof. We give the data and the exact index argument. The executable verification is reproduced in the supplement as `src/verify_quintic.m`. It uses a fresh Magma process and explicitly sets every class-group proof bound to the Minkowski bound [4]. It obtains $\mathrm{disc}(K) = 11^4$ and class number 1. No conditional class-group setting is used.

The complete torsion subgroup is $(\mathbb{Z}/2\mathbb{Z})^2$. The routine `PseudoMordellWeilGroup` returns `true`, together with a subgroup H of finite odd index in $E(K)$, generated by

$$\begin{aligned} T_1 &= (-r^4 + 5r^2 + r - 6,\ 0), \\ T_2 &= (-2r^4 + 2r^3 + 9r^2 - 7r - 6,\ 0), \\ P &= (7r^4 + r^3 - 22r^2 - 5r + 11, \qquad 8r^4 - 19r^3 - r^2 + 31r - 23). \end{aligned}$$

The abstract group returned is $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}$, with these generators in that order. The two torsion points generate all torsion. The computation also checks that the 2-Selmer group has order 8 and $2P \neq O$. Since all torsion has exponent 2, P has infinite order; the Selmer bound gives rank at most $3 - 2 = 1$. Thus $E(K)$ has rank one.

The call `Saturation([P],5 : TorsionFree:=true)` returns P itself. This certifies saturation through primes 2, 3 and 5, with torsion generators omitted from the returned list. Because the full torsion subgroup has order 4, the unchanged generator shows that 3 and 5 do not divide $[E(K) : H]$. Explicitly, for an odd prime ℓ, divisibility of the image of P in $E(K)/E(K)_{\mathrm{tors}}$ can be lifted to divisibility of P in $E(K)$: multiplication by ℓ is invertible on the torsion subgroup. Thus the saturation check and the odd-index assertion give

$$\gcd([E(K) : H], 30) = 1. \tag{7}$$

At the prime 3 the curve and the map have good reduction. The exact norms are

$$N_{K/\mathbb{Q}}(\mathrm{disc}\, E) = 2^{20}11^{12}, \qquad N_{K/\mathbb{Q}}(b) = 11^3.$$

Magma's elliptic Chabauty call, with the subgroup map and τ, returns the following data:

Quantity	Returned value
Prime	3
Upper bound in H	2
Points found	$T_2 - P,\ T_2 + P$
Index coprimality integer	15
Images under τ	$\infty,\ \infty$

The documented contract of this routine [3] makes the same upper bound valid in $E(K)$ when its subgroup index is coprime to the returned integer. Equation (7) verifies this hypothesis. Hence all rational-image points have been found. Exact elliptic addition identifies $T_2 \pm P$ with (6), completing the classification. □

Proof of Theorem 1. The polynomial f is irreducible, as shown above. A rational affine substitution preserves irreducibility because it is an automorphism of $\mathbb{Q}[x]$. If a rational quadratic substitution made f reducible, Lemma 1 would give a point of $v^2 = q(t)$ with finite rational t. The explicit inverse map in Section 3 would give a point of $E(K)$ with finite rational τ, contradicting Proposition 1. This covers all substitutions in the definition. □

5 Reproduction and verification

The supplement has two complementary verification programs:

- `src/verify_quintic.m` checks the field, model identity, full torsion, finite odd index, saturation, good reduction, and the Chabauty output. The complete successful output is supplied. The calculation was run with Magma V2.30-1.
- `src/verify_model.py` uses Python's standard-library rational arithmetic. It independently checks irreducibility modulo 2, the root-field representation, the quartic identity, the named points on the elliptic curve, and the identification of $T_2 \pm P$ with the two poles.

Run the Magma file in a fresh session. Run the Python check with `python src/verify_model.py`. The final Magma output reports an upper bound of 2, index condition 15, and two images equal to infinity. All assertions must pass. The accompanying README gives the public calculator option and records the tested software version.

The global point classification is the computer-assisted part of the proof. The separate Python calculation verifies its algebraic connection to the polynomial problem and the explicit points; its arithmetic does not depend on Magma.

References

[1] L. Du, *2-Superirreducibility of Univariate Polynomials over* $\mathbb{Q}$ *and* $\mathbb{Z}$, arXiv:2409.16206v2, 17 June 2025. arxiv.org/abs/2409.16206.

[2] A. Sahay, answer to *Composing an irreducible polynomial with a quadratic polynomial*, MathOverflow, updated 1 June 2025. mathoverflow.net/questions/495180.

[3] Computational Algebra Group, University of Sydney, *Magma Handbook: Curves over Number Fields; Mordell–Weil Groups and Descent Methods*, sections on elliptic Chabauty and saturation. Elliptic Chabauty documentation; descent and saturation documentation.

[4] Computational Algebra Group, University of Sydney, *Magma Handbook: Ideal Class Groups*, section on proof bounds. Class-group documentation.